Resilience-Aware, Human-Centric AI-RAN Orchestration for Service Continuity Across Terrestrial and Non-Terrestrial Networks

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Abstract—We describe a methods-ready research artifact for privacy-preserving, phone-first field measurement and twin-informed AI-RAN orchestration under heterogeneous terrestrial and non-terrestrial connectivity. The artifact couples canonical JSON Schema contracts, an integrated Gate 2 pipeline, preregistered evaluation design, and a 54-cell controlled pilot matrix (3 zones × 2 network conditions × 3 days × 3 workload profiles). We explicitly disclaim carrier-grade AI-RAN, deployable 6G infrastructure, citywide generalization, and unauthorized RF transmission. Empirical pilot results are RESULTS_PENDING_AUTHENTIC_GATE3_DATA; no outcome numbers are reported here.

Index Terms—6G-aligned, IMT-2030, AI-RAN, O-RAN, digital twin, non-terrestrial networks, reproducibility, privacy-preserving measurement

I. Introduction

Sparse, degraded, and disrupted connectivity remains a practical barrier to service continuity for learning, creation, and sensing workloads in under-connected communities. IMT-2030-aligned research explores AI-native air interfaces, open RAN disaggregation, and multi-access resilience including non-terrestrial network (NTN) fallback [1]–[3]. Operational claims about carrier-grade AI-RAN or deployable 6G infrastructure are outside the scope of this artifact.

This paper documents the methods-ready umbrella repository gunnchos-7gc-ai-ran-field-kit, which locks a doctoral thesis hypothesis and three research questions (RQ1–RQ3) in Gate 1, provides executable contracts and orchestration in Gate 2, defines controlled evidence collection in Gate 3, and preregisters scientific evaluation in Gate 4. We separate engineering completion from release evidence: Gate 2 system pass does not imply immutable archive, DOI deposit, or non-author reproduction.

A. Central hypothesis

A privacy-preserving orchestration architecture that combines physical device measurements, workload and service intent, digital-twin context, and terrestrial, local-edge, and NTN state can improve service continuity and worst-user quality of experience under sparse connectivity when compared with static, network-only, and service-priority baselines, while maintaining explicit fairness, energy, privacy, and reliability constraints.

B. Research questions

RQ1 (Trustworthy context): How can privacy-preserving Edge-IO measurements be transformed into reproducible digital-twin state without direct identifiers? RQ2 (Human-centric orchestration): To what extent does twin-informed AI-RAN orchestration improve continuity and worst-user QoE versus defined baselines? RQ3 (Multi-access resilience): Under which outage, latency, and privacy conditions do terrestrial, offline, and NTN fallback policies help or fail?

C. Contributions (methods only)

- Canonical cross-repository JSON Schema contracts and provenance for Edge-IO → twin → AI-RAN → NTN resilience paths.
- A preregistered statistical analysis plan with locked primary outcome recovery_time_s and baseline registries frozen before authentic pilot inspection.
- A 54-cell pilot design matrix with explicit consent, privacy scan, and session-mode rules separating calibration, rehearsal, and eligible pilot sessions.
- Honest claim boundaries: synthetic Gate 4 infrastructure results are labeled separately from pending Gate 3 field evidence.

Outcome tables and effect sizes are withheld until authentic Gate 3 pilot data exist (RESULTS_PENDING_AUTHENTIC_GATE3_DATA).

II. Related Work

A. IMT-2030 and 6G vision

ITU-R Recommendation M.2160 defines the framework and overall objectives for IMT-2030 development [1]. Survey literature outlines AI-native air interfaces, extreme connectivity, and integrated sensing and communication as research directions rather than deployed capabilities [2], [3]. We use 6G-aligned and IMT-2030-aligned wording only.

B. Open RAN and AI-RAN experimentation

The O-RAN Alliance architecture disaggregates the RAN into open interfaces suitable for policy and ML experimentation [4]. This artifact implements AI-assisted

TABLE I
Research gates and evidentiary meaning

Gate	Meaning
Gate 1	Locked thesis, hypothesis, RQ1–RQ3
Gate 2	Executable schemas + integrated pipeline
Gate 3	Authentic controlled-device pilot evidence
Gate 4	Preregistered evaluation on eligible evidence
Gate 5/6	Release archive, DOI, non-author reproduction

O-RAN-style control experiments in simulation; it does not claim near-real-time RIC deployment on commercial networks.

C. Edge computing and measurement

ETSI MEC defines a framework for edge-hosted applications and orchestration [5]. Companion work in edge-io-measurement-node specifies opt-in, schema-validated telemetry without payload capture; we reference that schema rather than redefining it here.

D. Non-terrestrial networks

3GPP TR 38.821 documents NR solutions for NTN scenarios including satellite and aerial platforms [6]. Our resilience baselines (terrestrial_only, always_ntn_on_terrestrial_failure, etc.) are evaluated in source-validated simulation paths; live NTN control is not claimed.

E. Reproducibility and preregistration

Preregistered outcomes, holdout registries, and negative-result preservation follow open-science reproducibility practices documented in evaluation/EVALUATION_PREREGISTRATION.md (formal preregistration citation: CITATION_NEEDED). This repository locks PRIMARY_OUTCOME_LOCK.json and EVALUATION_PREREGISTRATION.md before complete pilot result inspection.

III. System Architecture and Contracts

A. Umbrella orchestration role

gunnchos-7gc-ai-ran-field-kit is the canonical contract authority linking:

- edge-io-measurement-node — privacy-preserving device telemetry (RQ1),
- 7gc-digital-twin — twin-state construction (RQ1–RQ2),
- spectrumx-ai-ran-gary — AI-RAN policy baselines and ablations (RQ2–RQ3),
- ntn-resilience-sim — multi-access resilience simulation (RQ3).

Optional PHY supporting studies (e.g., ReadyGary beam selection) may inform assumptions but are not required for thesis completion.

B. Gate model

C. Canonical schemas

Contracts under contracts/ include measurement session context, Edge-IO batches, twin bundles, AI-RAN decision bundles, and Gate status reports. All integrated runs emit SHA-256 provenance via scripts/verify_provenance.py. Invalid fixtures under fixtures/invalid/ document rejection reasons for contract tests.

D. Integrated pipeline

The default reproduction path executes:

```
make test && make integrated-pipeline \
 EDGE_INPUT=fixtures/valid/edge_measurement_batch.valid.json
```

Outputs land under results/integrated/ with evidence labels synthetic or controlled_device_measurement. Synthetic outputs support Gate 4 infrastructure readiness only; they must not be presented as field pilot results.

IV. Measurement and Pilot Protocol

A. Evidence tiers

We distinguish smoke CI tests, synthetic reproducible benchmarks, controlled device measurements, and operational RAN claims (the last is not claimed).

B. Session modes

Only sessions with session_mode=PILOT count toward the 54-cell matrix after consent, privacy scan, and assignment-hash verification. Calibration and rehearsal sessions support infrastructure validation and are explicitly excluded from pilot counts.

C. Pilot matrix design

The matrix spans:

- 3 zones (zone_a, zone_b, zone_c) with public labels pending human approval,
- 2 network conditions (wifi_normal, wifi_degraded) applied via scripted condition scripts,
- 3 collection days (day_01–day_03),
- 3 workload profiles (learn, create, sense), each 300 s.

Authoritative cell definitions live in pilot/54_CELL_ASSIGNMENT_MATRIX.csv. At manuscript freeze: 0/54 eligible cells collected; HUMAN_ACTION_REQUIRED.

D. Workload protocols

Each workload specifies periodic request cadence, timeout handling, and success criteria (duration $\geq$300 s, samples $\geq$50, explicit nulls for unavailable metrics). Details are in pilot/WORKLOAD_PROTOCOLS.md.

E. Privacy and ethics

Collection follows opt-in consent, no payload inspection, no direct identifiers, aggregated public reporting, and receive-only RF boundaries where SDR observation is discussed (see docs/PRIVACY_AND_ETHICS.md; formal IRB citation: CITATION_NEEDED). Raw device exports remain in gitignored datasets/controlled/raw-private/; sanitized derivatives require human approval before publication.

F. Network condition protocol

Condition application and restoration use pilot/scripts/apply_condition.sh, verify_condition.sh, and restore_condition.sh with parameters in pilot/configs/network_conditions.yaml. Deviations are logged in datasets/controlled/protocol-deviations/PROTOCOL_DEVIATION_LOG.md.

V. Evaluation Design and Statistical Plan

A. Primary claim and outcome

The preregistered primary claim states that twin-informed service-aware orchestration reduces service outage or recovery time relative to static, network-only, and service-priority policies under defined degraded connectivity, subject to energy, fairness, privacy, and reliability constraints. The locked primary outcome is recovery_time_s (SHA-256 lock in evaluation/PRIMARY_OUTCOME_LOCK.json).

B. Baselines

AI-RAN baselines: static_uniform, network_only, service_priority, optimization_based, and proposed twin_informed. Resilience baselines include terrestrial_only, terrestrial_then_offline, always_ntn_on_terrestrial_failure, priority_class_fallback, service_aware_multi_access, and analysis-only oracle_hindsight. Registries: evaluation/BASELINE_REGISTRY.yaml, evaluation/ABLATION_REGISTRY.yaml, evaluation/HOLDOUT_REGISTRY.yaml.

C. Holdouts and ablations

Leave-one-out splits by day, zone, network condition, and workload profile are pre-defined under evaluation/splits/ and Gate 4 run artifacts. A stress-scenario holdout JSON specifies scenarios excluded from primary fitting. Split leakage checks and duplicate detection are mandatory before any primary-outcome inference.

D. Statistical safeguards

Analysis follows evaluation/STATISTICAL_ANALYSIS_PLAN.md: grouped summaries by day and zone; confidence intervals and effect sizes; practical significance thresholds from PRACTICAL_SIGNIFICANCE_THRESHOLDS.yaml; missing-data and protocol-deviation audits; cluster-aware treatment of repeated measures. The 54 sessions are not treated as 54 independent participants.

E. Synthetic Gate 4 infrastructure

Gate 4 automation demonstrates evaluation plumbing on synthetic or fixture-backed inputs. Those runs are labeled GATE4_EVALUATION_READY infrastructure, not authentic pilot completion. No numeric results from Gate 4 are imported into this methods manuscript.

VI. Results

RESULTS_PENDING_AUTHENTIC_GATE3_DATA

Authentic controlled-device pilot sessions required for primary-outcome analysis have not been collected at manuscript freeze. The eligible pilot matrix status is 0/54 cells. Calibration evidence (e.g., one valid Pixel 6a calibration session) supports infrastructure validation only and must not be used to claim pilot completion or causal superiority.

When authentic Gate 3 data exist, this section will report:

- pilot coverage against pilot/54_CELL_ASSIGNMENT_MATRIX.csv,
- primary outcome recovery_time_s by baseline and holdout split,
- uncertainty intervals, effect sizes, and practical significance against locked thresholds,
- negative and neutral results per evaluation/NEGATIVE_AND_NEUTRAL_RESULTS.md,
- failure boundaries per evaluation/FAILURE_BOUNDARIES.md.

No tables, figures, p-values, or effect sizes are reported here to avoid inventing empirical numbers.

VII. Limitations and Threats to Validity

A. Scope limits

This artifact is a research prototype, not commercial 6G, carrier-grade AI-RAN, or citywide deployment infrastructure. Gary/NWI anchor context supports digital-equity motivation but does not justify global generalization.

B. Evidence limits

Synthetic benchmarks and Gate 4 infrastructure runs demonstrate reproducible engineering and evaluation plumbing only. Until Gate 3 authentic pilot data exist, all outcome claims remain RESULTS_PENDING_AUTHENTIC_GATE3_DATA.

C. Design threats

Repeated measures within days and zones induce clustering; the statistical plan addresses this but cannot remove locale-specific confounds. Network condition scripts approximate degradation; they do not emulate full RF propagation diversity. NTN resilience paths are simulation-backed unless and until validated against controlled NTN-capable environments.

D. Privacy and consent

Public release depends on sanitized derivatives and consent-summary approval. Missing cells, protocol deviations, and consent withdrawals must be reported rather than imputed without documented sensitivity analysis.

E. Release and reproduction

Clean-checkout author reproduction, non-author reproduction, immutable release tarball, and DOI deposit remain Gate 5/6 requirements and are pending at manuscript freeze.

VIII. Conclusion

We presented a methods-ready, honesty-bounded artifact for resilience-aware, human-centric AI-RAN orchestration research spanning privacy-preserving measurement, twin-informed policy evaluation, and NTN fallback simulation. Gate 1 thesis elements, Gate 2 contracts, Gate 3 pilot protocol, and Gate 4 preregistration are in place; empirical pilot outcomes remain RESULTS_PENDING_AUTHENTIC_GATE3_DATA.

Future work completes the 54-cell matrix under human-approved zones and dates, executes the locked statistical analysis plan on eligible evidence only, and pursues release/archive steps with independent reproduction before outcome claims appear in a results manuscript.

References

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Appendix A
Reproducibility Appendix

A. Environment

See repository ENVIRONMENT.md for Python version, dependency pins, and sibling repository layout under REPOS_ROOT.

B. Author reproduction (pending)

AUTHOR_REPRODUCTION_REPORT.md records a clean-checkout run outcome; status at freeze: PENDING.

C. Non-author reproduction (pending)

NON_AUTHOR_REPRODUCTION_REPORT.md is an authentic empty pending template until an independent replicator files results.

D. Core commands

```
pip install -r requirements.txt
make test
make contract-test
make integrated-pipeline \
 EDGE_INPUT=fixtures/valid/edge_measurement_batch.valid.json
make gate1-validate
```

E. Pilot collection (human-gated)

Pilot counting requires pilotctl.py workflows, approved zone/date tokens, and storage under datasets/controlled/raw-private/ (gitignored). Public artifacts document protocol only until sanitized publication approval.

F. Build status

PDF build uses pinned Tectonic (scripts/build_paper.sh, version pinned in-script) or a digest-pinned TeX Live Docker image; local pdflatex/bibtex remains an optional fallback. See paper/Makefile and paper/README.md.